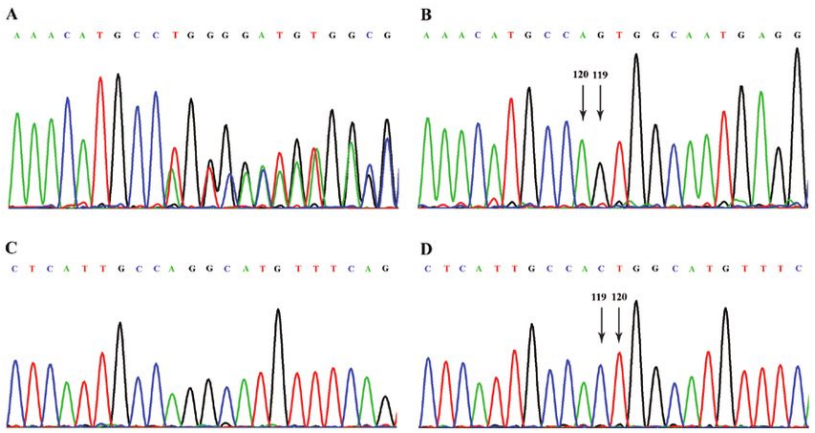
BME 105: Genetics in the Genomics Era

Lecture 4: Identifying genetic variants, part 1 (Sanger seq)



Instructor: Dr. Shelbi Russell,
Assistant Professor

Course website on canvas.ucsc.edu

Some slides modified from Angela Brooks

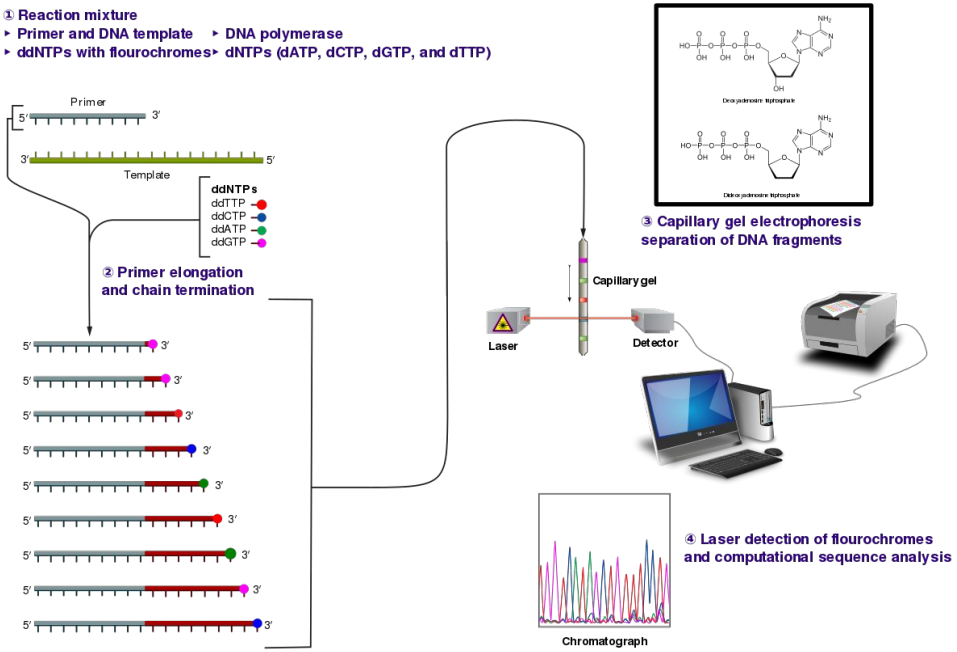
Lecture 4 Learning Objectives

- Become familiar with the types of DNA sequencing used for genome assembly (pre-2000):
- Be able to describe the steps in the polymerase chain reaction (PCR)
  - and how its used in genotyping
- Be able to articulate the similarities and differences between the chain-termination method and short-read sequencing by synthesis (i.e., Illumina sequencing)
- Understand why the chain-termination method (i.e., Sanger sequencing) is lower-throughput than short and long read technologies
  - Imagine the effort required to clone the entire human genome a few thousand bp at a time
- Be able to articulate what "shotgun" means in terms of genome sequencing
- Learn how genetic variants are categorized (SNP/SNVs, indels, CNVs, etc.)
- Be able to read a Sanger sequencing chromatogram and detect genetic variants
- Be able to describe how SNPs can be genotyped by microarray technology

Roadmap: DNA sequencing and analysis for genetic variant identification

First-generation DNA sequencing (1977)

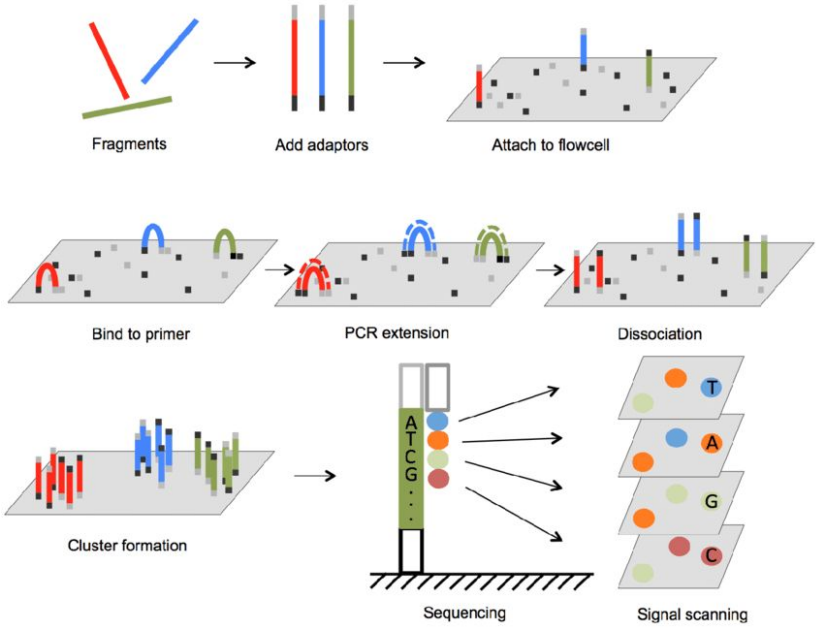
- Chain-termination method
- Sanger sequencing



Roadmap: DNA sequencing and analysis for genetic variant identification

First-generation DNA sequencing (1977)
- Chain-termination method
- Sanger sequencing

Second-generation DNA sequencing (~2000s)
- Massively parallel → high throughput
- One main platform until recently
  - Illumina sequencing (acquired Solexa)
  - [Roche 454 pyrosequencing - obsolete]
  - Illumina's main competitor: MGI



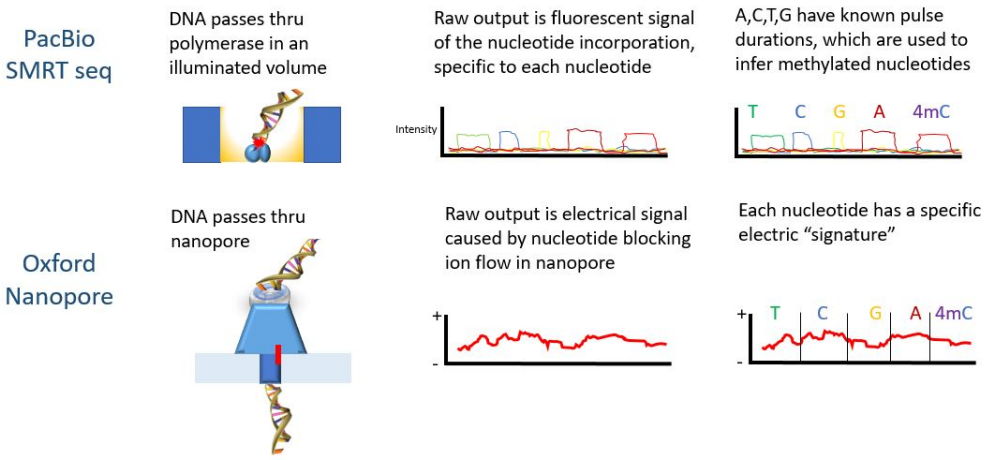
Illumina Sequencing

Roadmap: DNA sequencing and analysis for genetic variant identification

First-generation DNA sequencing (1977)
- Chain-termination method
- Sanger sequencing

Second-generation DNA sequencing (~2000s)
- Massively parallel → high throughput
- Illumina sequencing

Third-generation DNA sequencing (~2010s)
- Single-molecule
- Two main platforms:
  - Oxford Nanopore Technologies
  - PacBio

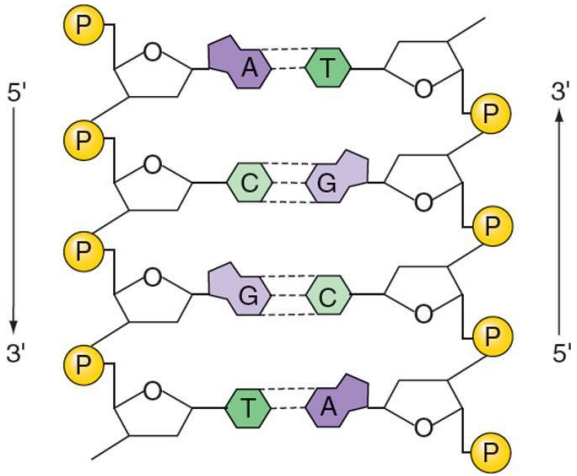


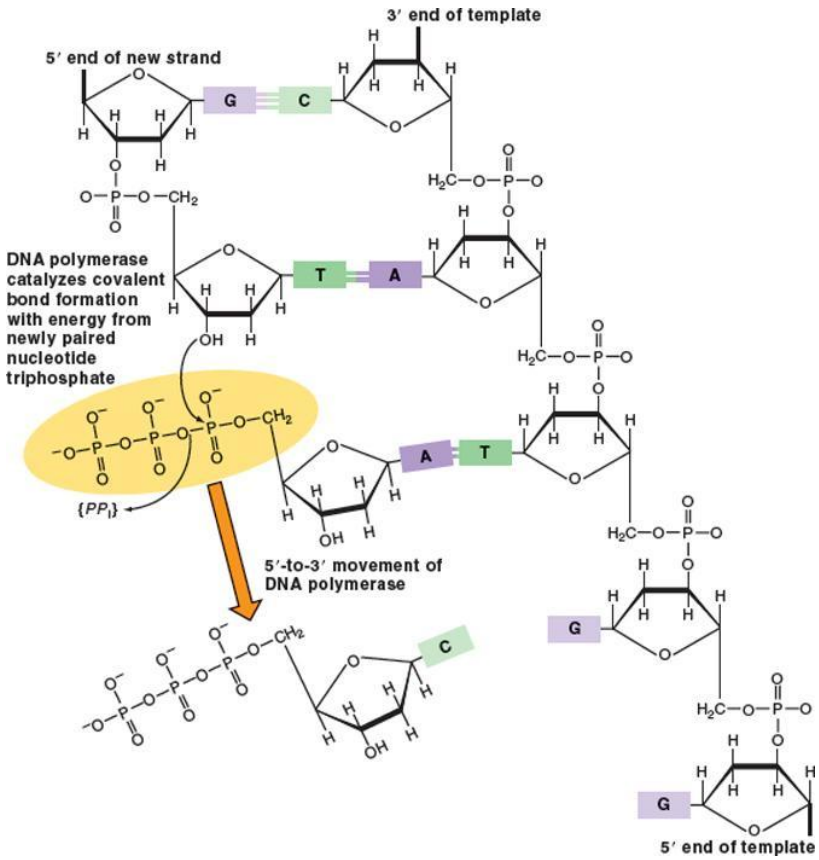
First-generation Sanger DNA sequencing

The chain-termination method

- Leverages DNA synthesis
- Requires PCR products as input

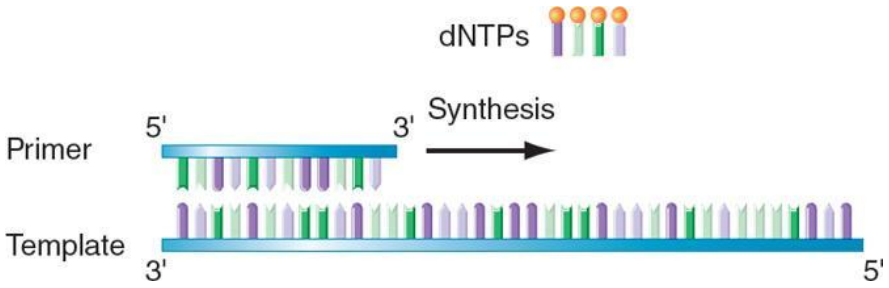
DNA base pairing

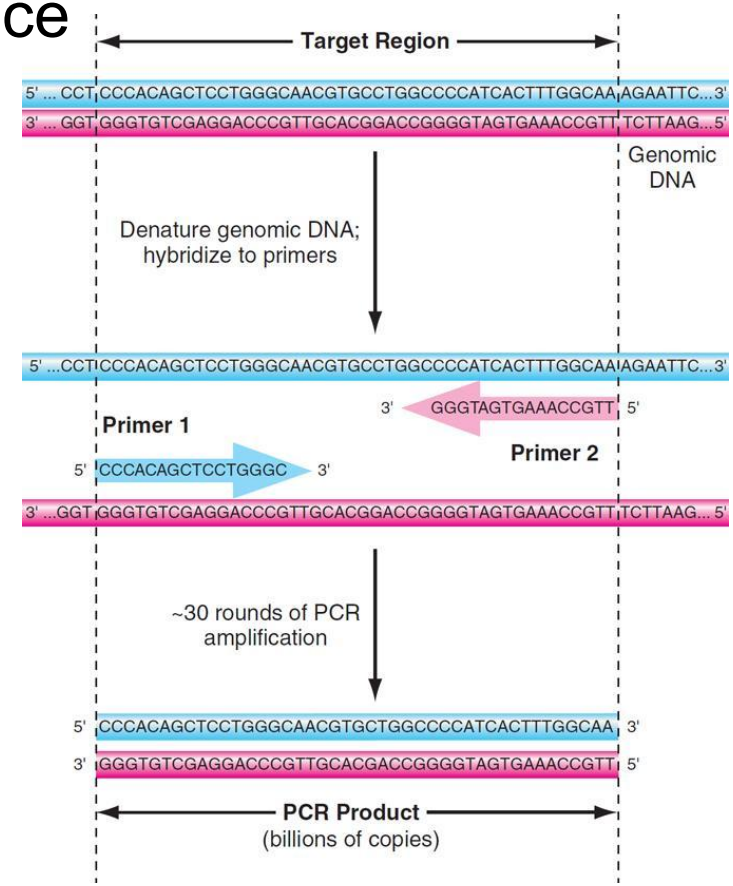




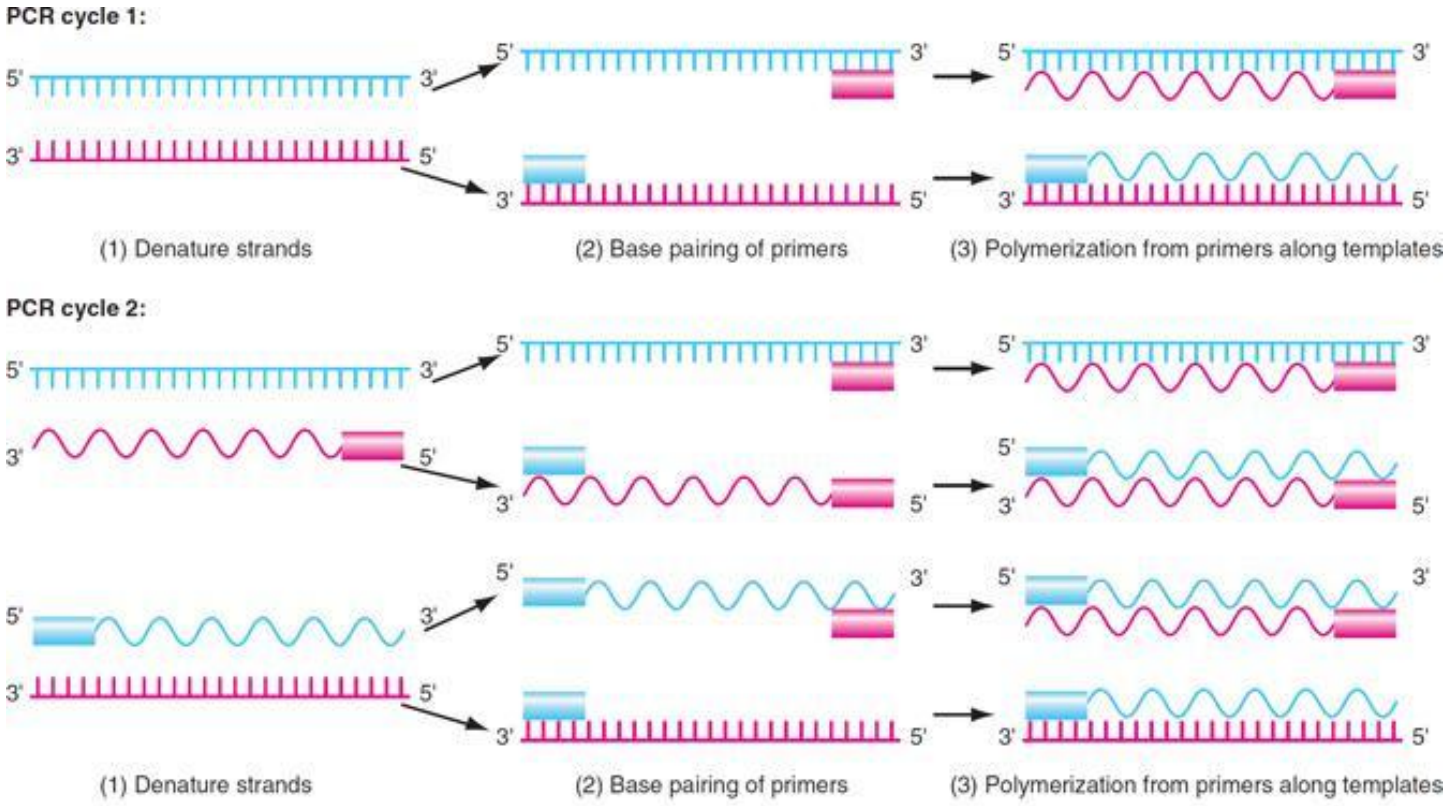
Polymerase chain reaction (PCR): Amplification of a target DNA sequence

PCR video: https://www.jove.com/science-education/10819/pcr

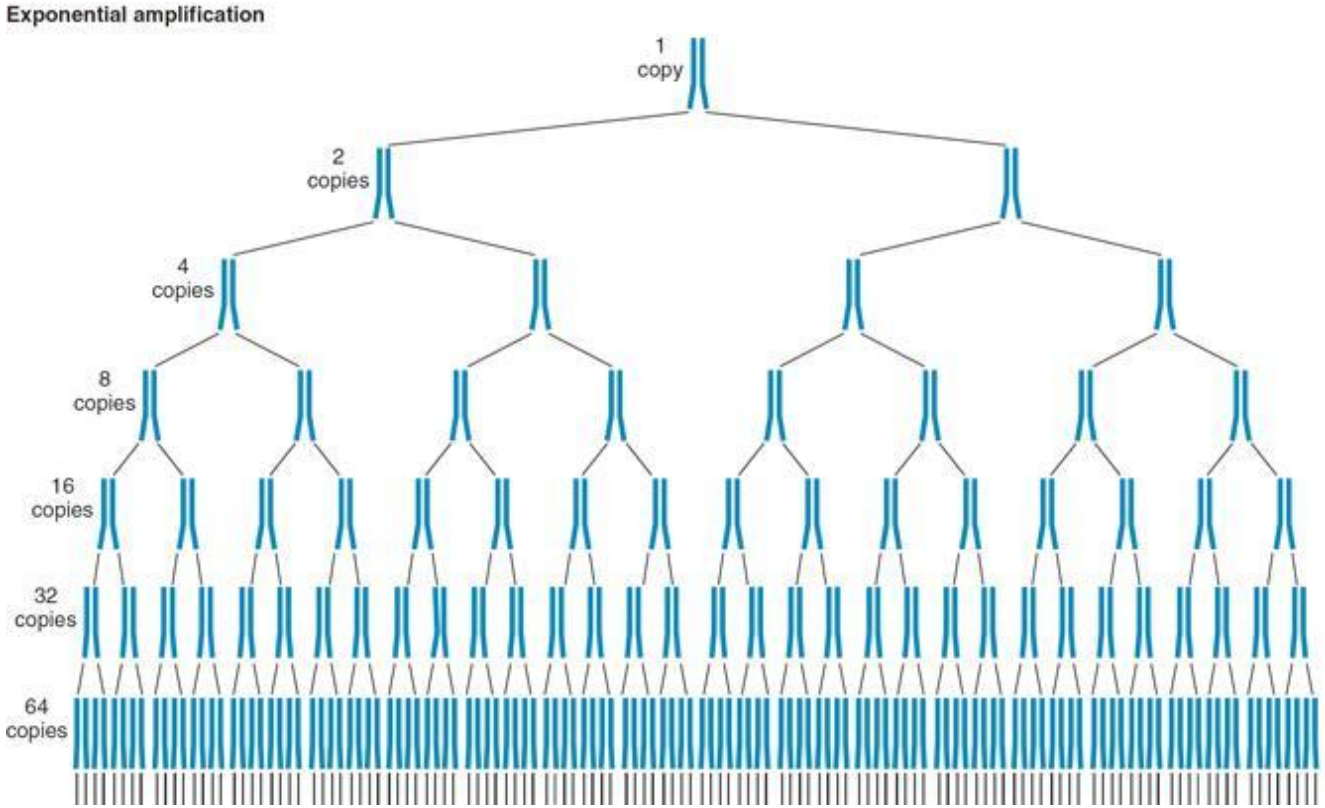




Polymerase chain reaction (PCR): Amplification of a target DNA sequence



Polymerase chain reaction (PCR): Exponential amplification of a target DNA sequence



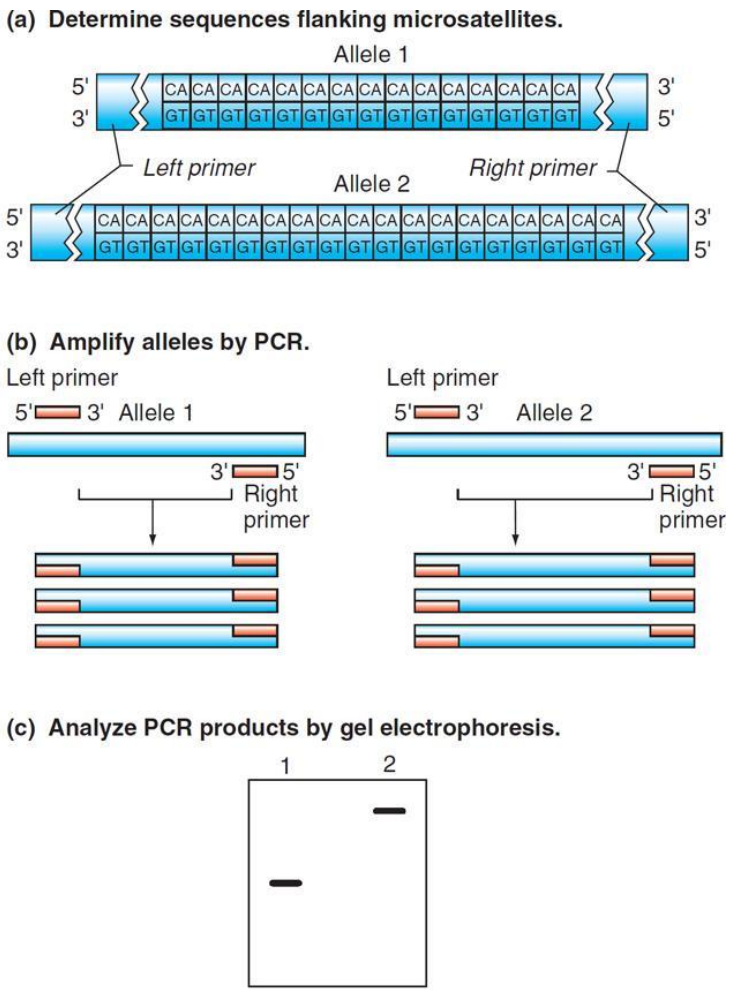
PCR alone can be used to detect allelic variation: microsatellites

SSR = single sequence repeat, also known as STRs = short tandem repeats

Microsatellites == SSRs

PCR primers in conserved regions, designed to straddle repeat polymorphism

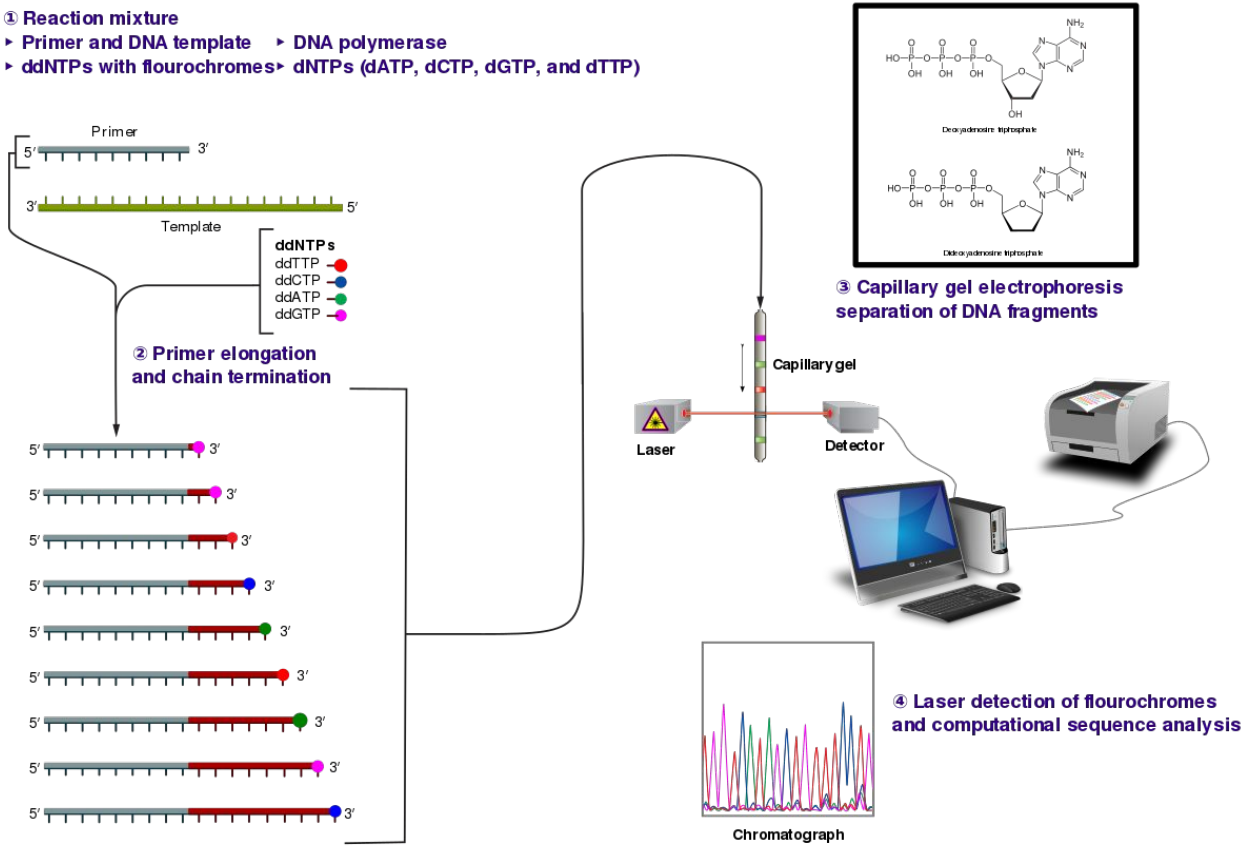
DNA amplicon length differences = repeat alleles



First-generation Sanger DNA sequencing

The chain-termination method

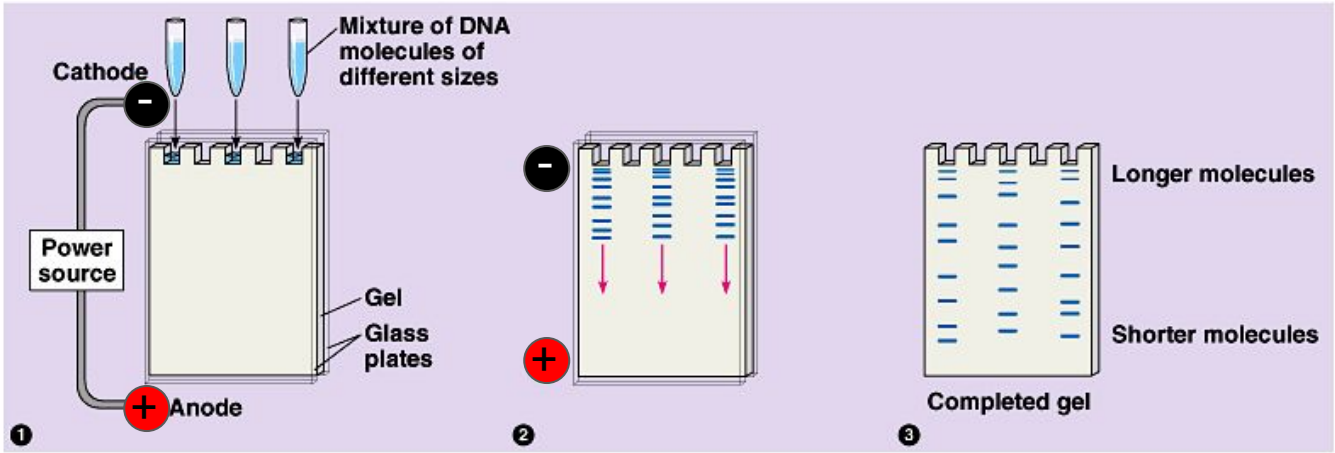
- Leverages DNA synthesis
- Incorporates fluorescent chain-terminator nucleotides to mark 3’ nucleotide
- Gel electrophoresis separates synthesis products by length
- Sequence is visualized by fragment order and color



Gel electrophoresis: Determine size of DNA molecules

- DNA is overall negatively charged due to the phosphate backbone
- Electric currents run negative to positive, or cathode to anode

→ You want to position the DNA with the cathode (-), so it runs to the anode (+)



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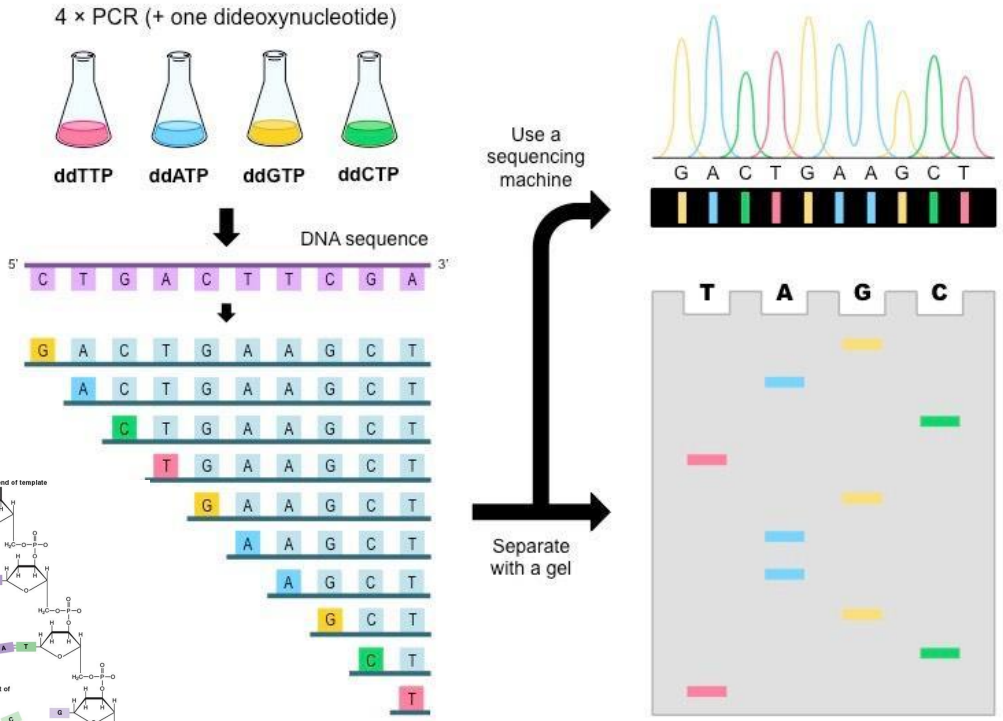
The DNA sequence is read in order by length and color

Sanger sequencing is similar to PCR, except you include a mix of dideoxynucleotides

- The sugar does not have an OH group, so the strand terminates

You obtain strands of different length

- The strand terminates at the included dideoxynucleotide sequence.



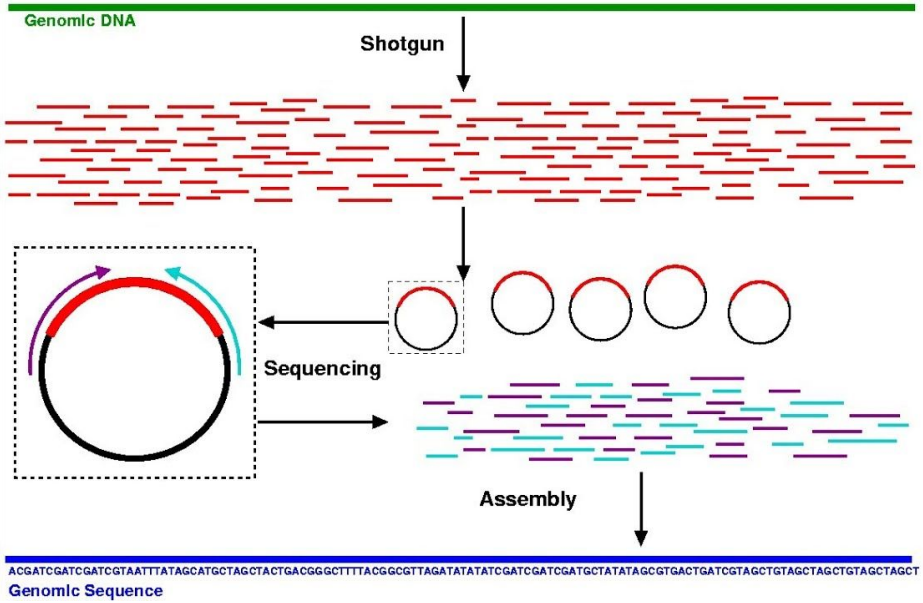
http://ib.bioninja.com.au

To assemble the human genome, a lot of cloning and computational biology was necessary

molecular cloning

1. **Shotgun sequencing** → random fragmentation of the genome for sequencing
2. Ligate fragmented genomic DNA (gDNA) into plasmid/fosmid/cosmid inserts
   a. Provides primer sites for sequencing
3. Grow ligated vectors in *E. coli* for purification and sequencing of pure traces
4. Repeat for the entire genome - must tile reads fairly deep across
5. Develop an assembly algorithm for overlapping sequencing reads and assembling contiguous tracts

*This was one of the major intellectual contributions from the Haussler lab at UCSC



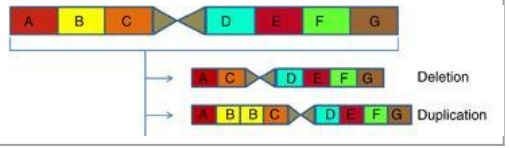
Categories of genetic variants

	Size	Frequency in human genome	Mechanism of generation	Example

SNP - Single nucleotide polymorphism	1bp	1 per 1kb	Mistakes in DNA replication; exposure to mutagens	`Allele 1 - CTGACTAGACGGC`
				`Allele 2 - CTGACAAGACGGC`
				`Allele 3 - CTGACCAGACGGC`

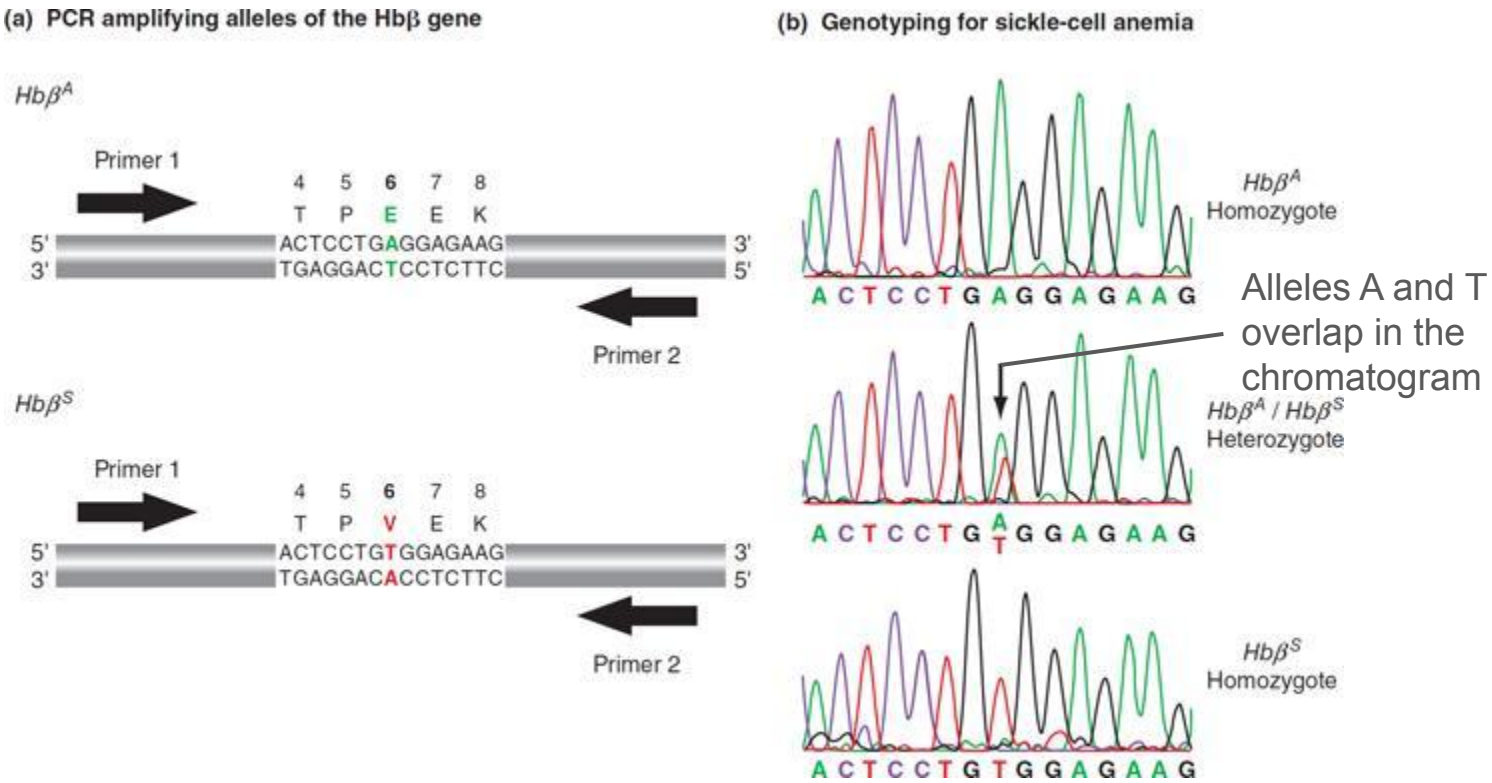
Indel - Insertion or deletion	~1-100bp	1 per 10kb	Mistakes in DNA replication (polymerase slippage - most common), recombination, and DNA damage repair	`Allele 1 - CTGACTAGACGGC`
				`Allele 2 - CTGAC-AGACGGC`
				`Allele 3 - CTGAC---ACGGC`

SSR - simple sequence repeat	1-10bp repeat unit	1 per 30kb	Once a repeat exists, easier to expand repeat due to slipped mispairing in DNA replication	`Allele 1 - CTGAGTGTGTGTCC`
				`Allele 2 - CTGAGTGTGTCCAG`
				`Allele 3 - CTGAGTGTCCAG`

CNV - copy number variant	10bp-1Mb	1 per 3Mb	Unequal crossing-over in Meiosis I	

How are genetic variants identified from Sanger reads?

SNPs:



Insertion-deletion (Indel) genotyping

- Indels change the sequence length
- In heterozygous samples, this shifts the alignment

Correct alignment

A

```
AAAGG-CAAGCAATAT
AAAGGGCAAGCAATAT
```

wrong alignment

B

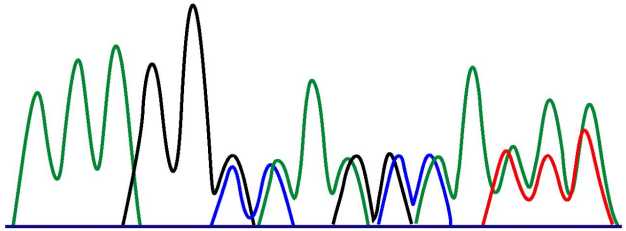
```
AAAGGCAAGCAATAT.
AAAGGGCAAGCAATAT
```

consensus

C

```
AAAGGSMARSMAWWW
```

D



deletion shifted the affected sequence

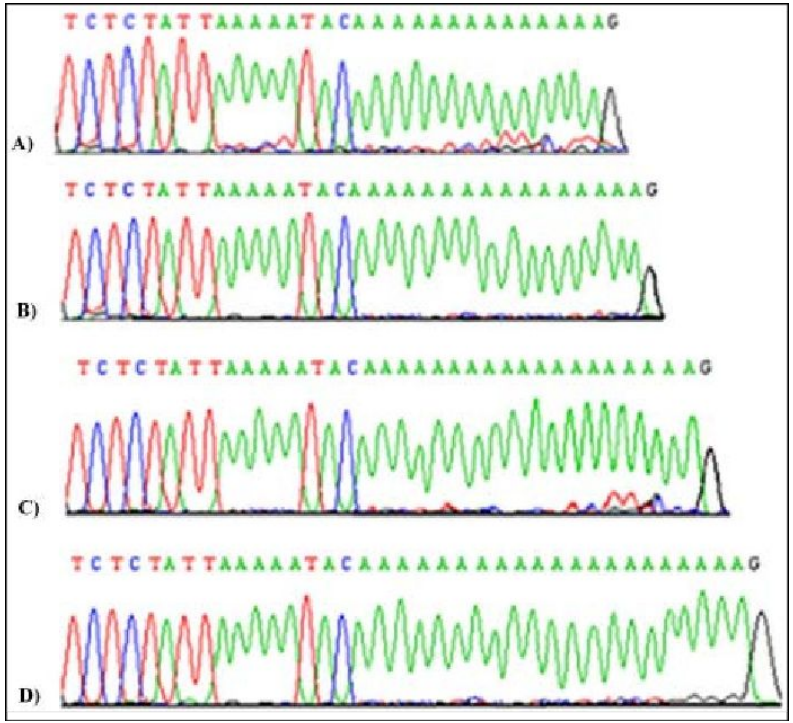
IUPAC symbols:

R = A or G
Y = C or T
K = G or T
M = A or C
S = C or G
W = A or T
B = not A
D = not C
H = not G
V = not T
N = any

Dmitriev and Rakitov 2008 PLoS Computational Biology

Simple sequence repeats (SSRs): microsatellite genotyping

- Also known as STRs (short tandem repeats)
- SSRs vary in length
- Consist of short repeating units
- Produced by slipped strand mispairing during DNA synthesis (aka replication slippage)



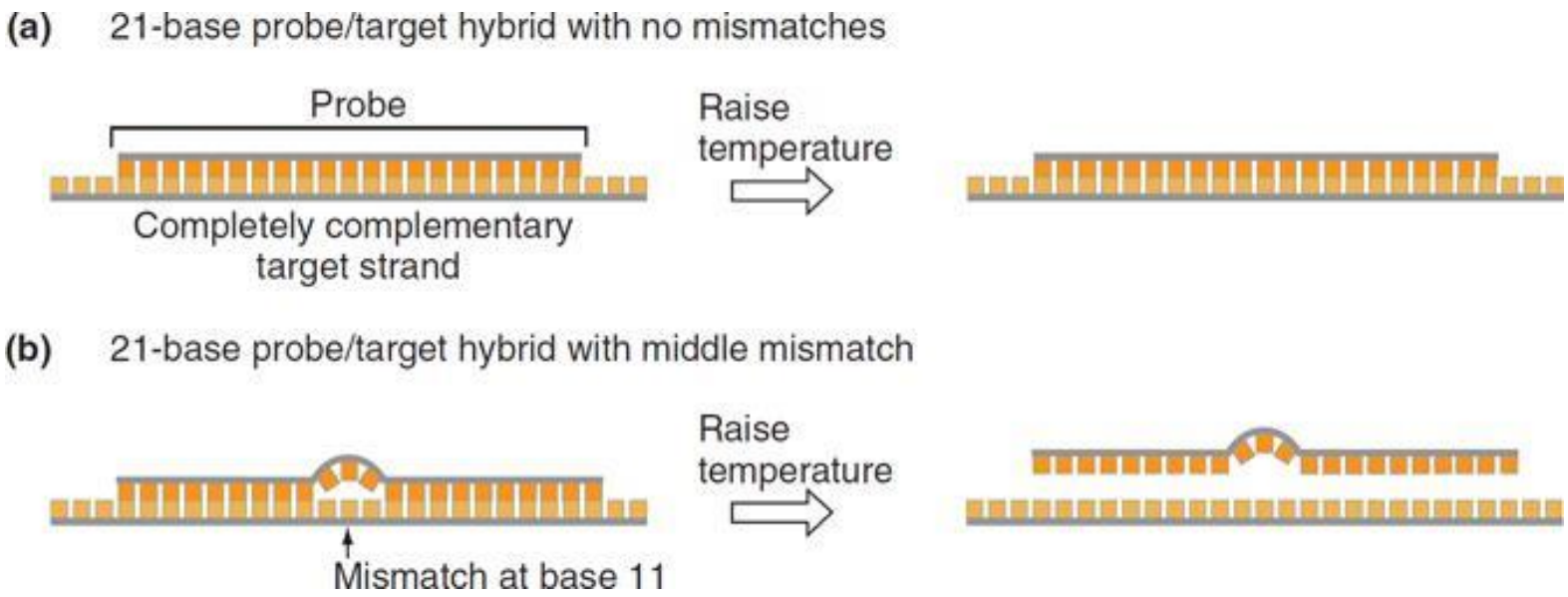
Colagar et al. 2016 Genetics in the 3rd Millennium

E.g., variation of poly(A) at the 3-end of the vitamin D receptor (VDR) gene

Genetic variant detection by SNP microarrays

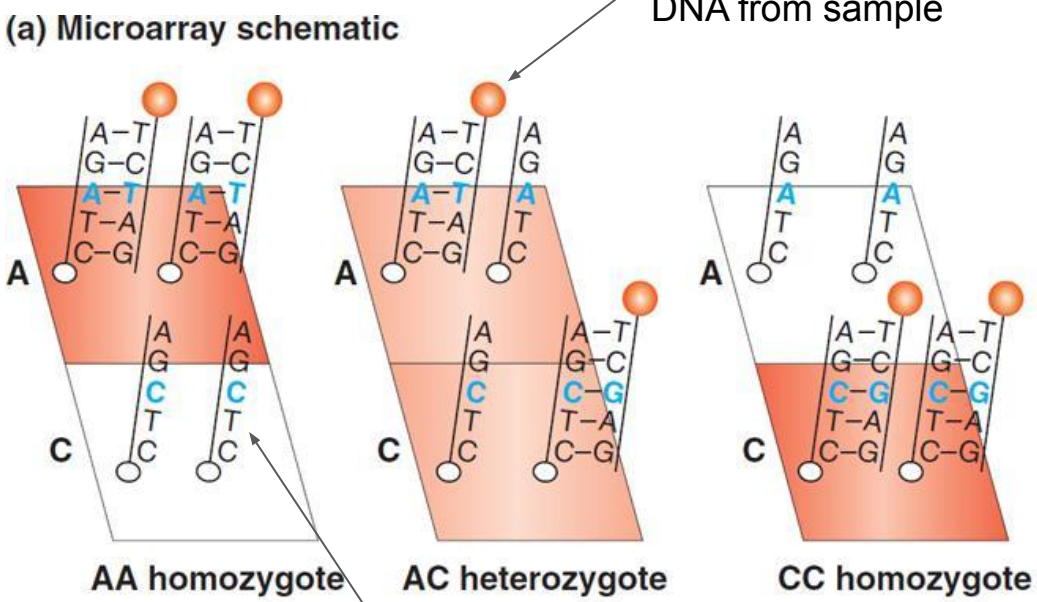
- Relatively cheap approach to genotype ~1 million SNPs, simultaneously
- Commonly used in large scale studies and cheaper commercial products (e.g. 23andMe, Ancestry)
- Consist of an array of DNA probes
  - Sequences complementary to genes/regions of interest
    - Disease markers
    - Markers correlated with ancestry
    - Other phenotypic traits

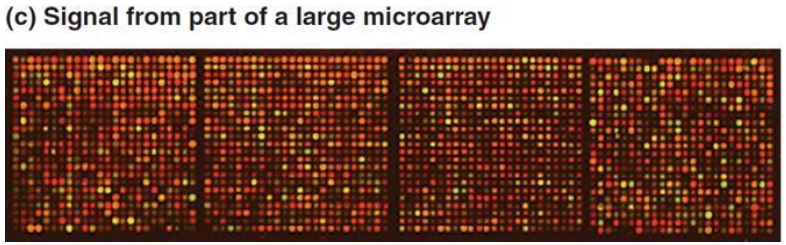
SNP microarray hybridization stringency



SNP microarrays

Fluorescently labeled
DNA from sample





**You have to know the sequence you want to print onto the microarray.

Pitfalls of Sanger sequencing and SNP microarrays

Must know primer/probe sequence

- Biased towards known sequences
- SNP arrays, sequence-specific primers, and vectors + primer sequences

Low-throughput

- Think of all the cloning required for unambiguous chromatograms
  - Across a complex genome (e.g., the human genome)
  - Across a diverse sample (e.g., the human microbiome)

- SNP chips only assay known variation (i.e., you must be able to design probes)

Chromatograms have limited resolution to disentangle mixed haplotypes

- Heterozygous samples
- Complex samples
- Contamination